Synthia: An Agentic AIOps Framework for Hybrid Multi-Cloud Orchestration with Predictive Telemetry and Human-in-the-Loop Supervision

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Abstract

The proliferation of microservices and hybrid multi-cloud deployments has exponentially increased the complexity of modern IT infrastructure. Traditional Artificial Intelligence for IT Operations (AIOps) platforms excel at aggregating logs and surfacing alerts but lack the capability to autonomously orchestrate remediation across varied environments without introducing significant security risks. This paper introduces Synthia, a self-hosted, Agentic AIOps framework designed to bridge the gap between predictive infrastructure observability and safe, automated remediation. By synthesizing a Bring-Your-Own-Key (BYOK) multi-cloud provider registry, a dual-engine speech-to-text pipeline for low-latency interactions, and a predictive machine learning anomaly detector, Synthia functions as an autonomous operational partner. To mitigate the "blast radius" inherent to Large Language Model (LLM) execution, Synthia strictly enforces a Human-in-the-Loop (HITL) mechanism. All destructive or state-mutating cloud operations proposed by the agent are securely sandboxed, mapped to specific provider SDKs, and must be manually approved by human engineers prior to execution. We present the system architecture, mathematical formulations for predictive failure, the HITL state machine, and simulated performance metrics demonstrating Synthia's efficacy in reducing incident resolution time while preserving operational security.

1 Introduction

Cloud computing architectures have transitioned from monolithic applications deployed on single virtual machines to distributed, containerized microservices orchestrated across multiple cloud vendors (e.g., AWS, Azure, Google Cloud) and on-premises environments. While this distributed approach offers unparalleled scalability and fault tolerance, it imposes a massive cognitive load on Site Reliability Engineers (SREs). When a service degrades, engineers must manually correlate metrics, traces, and log streams across disparate platforms to identify root causes, subsequently executing manual terminal commands or API calls to remediate the issue.

Recent advancements in Large Language Models (LLMs) have catalyzed the emergence of Agentic AIOps—systems that not only observe infrastructure but actively interact with it. However, granting an autonomous AI agent direct write-access to production environments introduces severe security vulnerabilities. An hallucinated command could inadvertently drop a production database, alter permissive Identity and Access Management (IAM) policies, or terminate critical container instances.

To resolve the tension between automation efficiency and operational safety, we developed **Synthia**. Synthia is a hybrid, multi-cloud orchestration engine that leverages voice-activated AI agents to manage cloud services. The primary contributions of this paper are:

1. **A Modular BYOK Architecture:** A decoupled control plane that allows users to seamlessly switch execution contexts between local Docker daemons, AWS, and Azure using a Bring-Your-Own-Key model.
2. **Predictive ML Telemetry Pipeline:** A hybrid model evaluating continuous metric vectors and log embeddings to predict node failures before they cascade.
3. **Strict HITL State Machine:** A mathematically defined execution loop that forces agentic LLMs to pause execution and seek explicit human authorization for state-mutating commands.
4. **Low-Latency Edge Voice Processing:** A client-side wake-word engine coupled with server-side `faster_whisper` transcription to enable hands-free incident management without compromising bandwidth.

2 Related Work

2.1 Traditional AIOps and Observability

First-generation AIOps platforms focus on anomaly detection and alert noise reduction. Systems utilize clustering algorithms (such as DBSCAN) to group related alerts and reduce alert fatigue for human operators. However, these systems are fundamentally reactive and advisory; they do not possess the orchestration capabilities to independently execute fixes across cloud boundaries.

2.2 LLM Agents in Software Engineering

The introduction of tools like Open-Interpreter and Kubiya has demonstrated the viability of utilizing LLMs to generate Infrastructure as Code (IaC) and execute terminal commands. While these frameworks are highly capable, they often operate in fully autonomous modes or rely on expensive, proprietary SaaS backends. Synthia diverges by emphasizing a self-hosted, open-architecture approach that integrates the open-source Model Context Protocol (MCP).

2.3 Human-in-the-Loop (HITL) Automation

Research into HITL systems emphasizes the necessity of human oversight in critical automation loops. In aviation and medical AI, HITL acts as a fail-safe. Synthia adapts these principles for cloud computing, treating infrastructure as a critical environment where the AI acts as a planner and the human acts as the ultimate execution authority.

3 System Architecture

Synthia is architected as a decoupled, multi-tier distributed system. The stack separates edge user interactions from computationally intensive backend NLP processing and cloud orchestration tasks.

3.1 Client Edge and Voice Pipeline

To enable seamless interaction during critical incident response, Synthia features a React-based frontend (built with Vite) that heavily utilizes the WebSpeech API. Unlike continuous-streaming assistants that consume high network bandwidth, Synthia utilizes a local wake-word engine (`useWakeWord.ts`). Audio is only captured and transmitted over the network once the local wake-word is verified. Upon activation, the audio payload is routed to a backend `faster_whisper` engine running on CTranslate2. If local compute is bottlenecked, the system automatically falls back to the OpenAI Whisper API.

3.2 FastAPI Core Control Plane and NLU

The brain of Synthia operates as a containerized FastAPI application. Once audio is transcribed to text, it passes through an intent classification layer. Synthia integrates with AWS Lex (`aws_lex.py`) to classify broad intents (e.g., "Check database status" vs. "Restart web container"). The intent is then passed to a custom Natural Language Understanding (NLU) Interpreter (`routing/interpreter.py`). The LLM generates a structured JSON payload representing the precise cloud API parameters required for the task.

3.3 Multi-Cloud Provider Registry

To prevent vendor lock-in and enable hybrid deployments, execution logic is abstracted via a Provider Registry pattern. When the interpreter determines the required action, the request is mapped to one of the following providers:

- **Docker Provider:** Communicates with the local or remote `/var/run/docker.sock` to inspect, start, stop, or prune containers.
- **AWS Provider:** Wraps the Boto3 SDK to manage EC2 instances, S3 buckets, and CloudWatch alarms.
- **Azure Provider:** Interfaces with Azure Resource Manager (ARM) APIs.

This layer enforces the BYOK security model. User API keys are injected dynamically per session, ensuring the FastAPI backend remains stateless and highly secure.

4 Methodology and Predictive Modeling

Synthia moves beyond reactive execution by proactively monitoring the infrastructure it manages. This is achieved via a background telemetry ingestion pipeline and a predictive machine learning model.

4.1 Telemetry Ingestion and Log Vectorization

Synthia ingests two data streams: structured metrics (CPU, RAM, Network I/O) and unstructured log streams. Let the system state at time t be represented by a continuous metric vector $\mathbf{X}_t \in \mathbb{R}^d$. Simultaneously, raw text logs are parsed into templates. A pre-trained embedding model maps the recent window of logs into a dense vector representation $\Phi(L_t) \in \mathbb{R}^k$.

4.2 Predictive Failure Formulation

To predict impending infrastructure failure (e.g., an Out-of-Memory kill or application deadlock), we utilize a hybrid classification model. The probability of failure within a future time horizon Δt is computed as:

$$P(F_{t+\Delta t} = 1 \mid \mathbf{X}_t, L_t) = \sigma\left(\mathbf{W}_m \mathbf{X}_t + \mathbf{W}_l \Phi(L_t) + b\right) \tag{1}$$

Where:

- $F_{t+\Delta t} \in \{0, 1\}$ is the binary indicator of failure.
- $\mathbf{W}_m \in \mathbb{R}^{1 \times d}$ represents the learned weights for the metric features.
- $\mathbf{W}_l \in \mathbb{R}^{1 \times k}$ represents the learned weights for the log embeddings.
- b is the bias term, and $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid activation function.

When $P(F_{t+\Delta t} = 1) > \tau$ (where τ is a configurable risk threshold), Synthia autonomously triggers the Reasoning Phase to draft a preemptive remediation plan.

5 Human-in-the-Loop (HITL) Execution Engine

The defining security feature of Synthia is its strict control over LLM execution capabilities. We formalize this using a discrete State Machine containing four states: $S_{observe}$, S_{reason}, S_{review}, and $S_{execute}$.

5.1 The 4-Stage State Machine

The execution pipeline ensures that no destructive action occurs without human consent.

1. $S_{observe}$ **(Telemetry):** The system passively ingests data. Transition to S_{reason} occurs if an anomaly is detected or a user issues a voice command.
2. S_{reason} **(Planning):** The LLM is invoked in a sandboxed environment to formulate an execution plan. The output is a declarative JSON structure outlining the exact SDK methods to be called.
3. S_{review} **(HITL Gate):** The system pauses and pushes an interactive approval card to the frontend. Transition to $S_{execute}$ strictly requires an explicit cryptographic authorization token from the human operator, $H_{auth} = 1$.
4. $S_{execute}$ **(Action):** The plan is routed to the Provider Registry and executed. The system monitors the return codes. If execution fails, the state transitions back to S_{reason} with the appended error logs for self-correction.

5.2 Blast-Radius Containment

Commands generated by the LLM are evaluated against an Open Policy Agent (OPA) ruleset before reaching S_{review}. Operations classified as high-risk (e.g., `docker system prune -a`, `aws ec2 terminate-instances`) trigger secondary multi-factor authentication requirements on the frontend.

6 System Implementation and Integration

6.1 Model Context Protocol (MCP)

To ensure Synthia remains extensible, the backend architecture exposes its internal tools via the Model Context Protocol (MCP). By running an MCP Server over standard input/output (STDIO) or Server-Sent Events (SSE), external agents (such as Claude Desktop or locally running LLMs via Ollama) can query Synthia's Provider Registry. This effectively turns Synthia into a universal operating system layer for AI applications.

6.2 Asynchronous Job Queues

Because LLM inference and cloud API executions can exceed standard HTTP timeout thresholds, Synthia utilizes an asynchronous background queue (via Inngest). The Next.js frontend immediately receives a `202 Accepted` response upon submitting a query, and subsequently subscribes to a Supabase real-time WebSocket to receive streaming updates as the execution progresses through the state machine.

7 Experimental Evaluation

To validate the architecture, we deployed Synthia in a simulated hybrid environment consisting of a local Docker Swarm (10 containers) and an AWS VPC containing 5 EC2 instances.

7.1 Latency Analysis

We measured the end-to-end latency from wake-word activation to the generation of the HITL approval card.

- **Local Wake-Word Detection:** $\sim$150 ms (executed entirely in-browser).
- **STT Transcription (`faster_whisper`):** $\sim$600 ms for a 5-second audio snippet.
- **LLM Reasoning and Planning:** $\sim$1.2 seconds (using optimized Groq/Llama-3 endpoints).

The total system response time consistently remained under 2.5 seconds, providing an interactive, conversational experience suitable for high-stress incident management.

7.2 Remediation Efficacy

In simulated failure scenarios (e.g., artificially introducing memory leaks to a Node.js container), the predictive telemetry pipeline successfully flagged anomalies 4.5 minutes before absolute container failure (OOM kill). The generated remediation plan (scaling the container memory limit and triggering a rolling restart) was accurate in 94% of test cases, requiring human operators only to click "Approve" to avert downtime.

8 Discussion and Future Work

The implementation of Synthia highlights the immense potential of pairing LLM reasoning engines with deterministic infrastructure SDKs. However, challenges remain in fine-tuning log embedding models to handle highly obfuscated or novel error stack traces.

Future development will focus on expanding the Provider Registry to include Kubernetes natively (via the `kubernetes-client`), allowing Synthia to orchestrate complex pod deployments and Helm chart rollbacks. Additionally, we plan to implement a Multi-Agent architecture where specialized sub-agents (e.g., a Security Agent, a FinOps Agent, and an Orchestration Agent) negotiate remediation plans collaboratively before presenting them to the human supervisor.

9 Conclusion

This paper presented Synthia, a comprehensive Agentic AIOps framework designed for hybrid multi-cloud orchestration. By decoupling the execution layer via a Provider Registry, employing proactive predictive machine learning, and enforcing a strict Human-in-the-Loop execution state machine, Synthia provides SREs with a powerful, autonomous assistant. The framework successfully mitigates the security risks associated with autonomous AI execution, ensuring that cloud environments remain secure while drastically reducing manual operational overhead.

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